Species Distribution Modelling (SDM) & R Data Wrangling

Honours Workbook & "Field" Guide

Weekly readings, concepts, and SDM practice

Clarke J.M. van Steenderen

clarke.vansteenderen@ru.ac.za | vsteenderen@gmail.com

Department of Zoology and Entomology, Rhodes University

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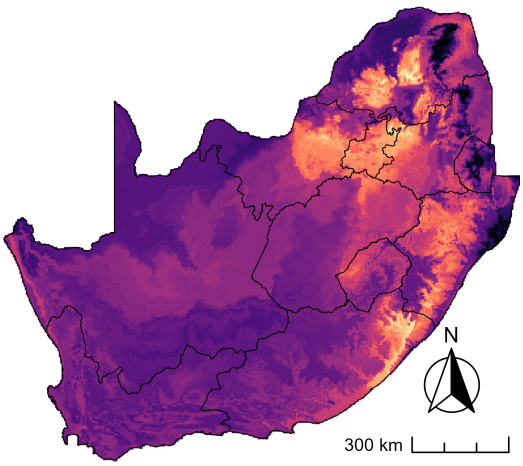

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1 Introduction

Goal: By the end of this module, you should be able to build, evaluate, and interpret species distribution models, and be confident in data manipulation and mapping in R.

All R scripts for this module are available at this GitHub repository

Species Distribution Models (SDMs) (sometimes referred to as species niche models, ecological niche models, or niche theory models) are statistical tools that predict the geographic areas where species might occur based on known occurrence records and environmental or other relevant variables. These predictions can be projected over time (e.g., using climate models to simulate future scenarios) and across different environments or spatial scales (e.g., specific areas within a country, entire provinces or districts, continents, or globally). Many SDM studies focus on invasive species to identify areas that might be climatically suitable for establishment and spread. Similarly, studies on threatened species aim to predict distribution changes under future climates, helping prioritise areas for conservation. SDMs are also increasingly applied in biological control programmes to assess where climatic conditions might support potential control agents.

The applications of SDMs are broad. Once you acquire the skills to build and evaluate your own models, you can create predictive maps for any taxon with available GPS records and environmental predictors. While the process may seem complex at first - especially tasks like model parameterisation or coding syntax - don't be discouraged. Keep experimenting, engaging with your classmates, and thinking about the new concepts. With practice, you will be able to confidently add SDMs into your research toolkit.

Recommended textbooks:

Bivand, R.S., Pebesma, E.J. and Gómez-Rubio, V., 2013. Applied spatial data analysis with R. New York, NY: Springer New York.

Franklin, J. and Miller, J.A., 2009. Mapping species distributions: spatial inference and prediction. Cambridge University Press.

2 Module outline

This Honours module will comprise class discussions every Friday based on prescribed readings, R activities, two group activities, and one final assignment that will count towards your class record.

Weeks 1 and 2 will drill R exercises that build up to implementing SDMs. Specifically:

Week 1:

1. Introductory lecture, and a mock debate for and against a housing development based on SDM output for an endangered frog
2. Data manipulation and summarising, writing and reading files, object types (e.g. lists, matrices, data frames, vectors, ggplots)
3. Downloading research-grade GPS records from GBIF
4. Plotting maps [with overlaid GPS points] using the ggplot package
5. Customising maps (e.g. adding scale bars, North arrow, legends)
6. A brief introduction to rasters, and a basic SDM exercise using a standard binomial GLM
7. Creating your own functions in R (very useful for repetitive tasks)
8. Working with lists and for-loops

Week 2:

1. Filtering and summarising large datasets (introduction to packages like `dplyr`)
2. extracting important variables and creating a variety of ggplots
3. An introduction to rasters and how to use them to store spatial information
4. Creating raster layers (multiple rasters into one)
5. The importance of raster resolution
6. Downloading current global temperature and precipitation raster layers from the WorldClim database and plotting them for a target area
7. Downloading future climate projections (2050, 2100 etc.) and comparing to the current climate

Weeks 3 and 4 will be dedicated to working through an SDM exercise for the Asian citrus psyllid (*Diaphorina citri*), and will cover:

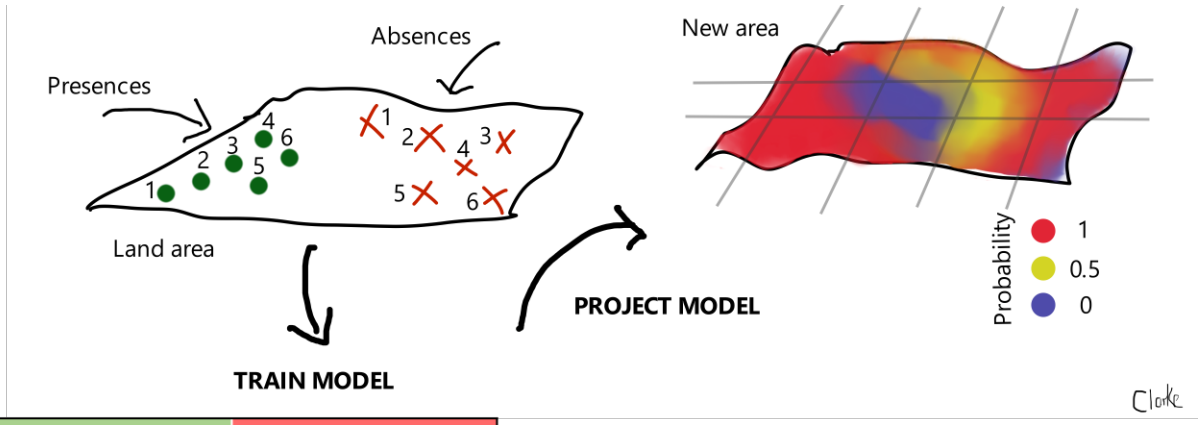
1. Variable selection (which climatic variables should be used to build a model?)
2. Background point generation (how do we create absence points?)
3. Data extraction (setting up the correct input data for the SDM workflow)
4. Model tuning (which parameters are optimal for the data?)
5. Ensemble approach to modelling (average over more than one model, including MaxEnt)
6. Projection and mapping (project the model onto a geographic area)
7. Output interpretation (which areas are most climatically suitable?)
8. GitHub basics: create a profile and repositories for code storage

3 What SDMs are not

Many people erroneously treat climatic suitability scores as direct probabilities of establishment. However, SDMs predict the relative climatic suitability of a particular site, reflecting the likelihood that the model will classify a species as present (scored as 1) or absent (scored as 0), based on environmental conditions (and/or other variables of interest). It is important to note that actual establishment depends on factors beyond climate alone, including biotic interactions, dispersal, habitat availability, and adaptive ability.

4 What SDMs are, and how they work

Suppose we have six confirmed GPS records for species A, and six sites where it does not occur (absence points), shown as green circles and red crosses in Figure 1. We aim to use average temperature and rainfall as variables (also termed covariates') to characterise this species' distribution. By extracting these data at each presence and absence point, we can train a predictive model to distinguish between presence (scored as 1) and absence (scored as 0).



	PRESENCES		ABSENCES	
Site	Rain	Temperature	Rain	Temperature
1	2.2	22	10.2	5
2	3.5	25	11.3	10
3	6	23	15.3	12
4	4.5	22	9.6	8
5	4.2	21	13.2	9
6	6.1	20	14.3	11
	1		0	

Figure 1: A simplified diagram explaining species distribution modelling.

If the model performs well, we can project it onto new sites, providing probability estimates of climatic suitability. The new area in Figure 1 represents a climatic suitability map, where

areas are coloured by probability, from 0 (least suitable; blue) to 1 (most suitable; red). The map suggests that if species A were to arrive in the new area, it would prefer the eastern and western regions, with the central areas being less suitable. At this point, you may have several questions about the model-building process and the reliability of these predictions.

Key questions include:

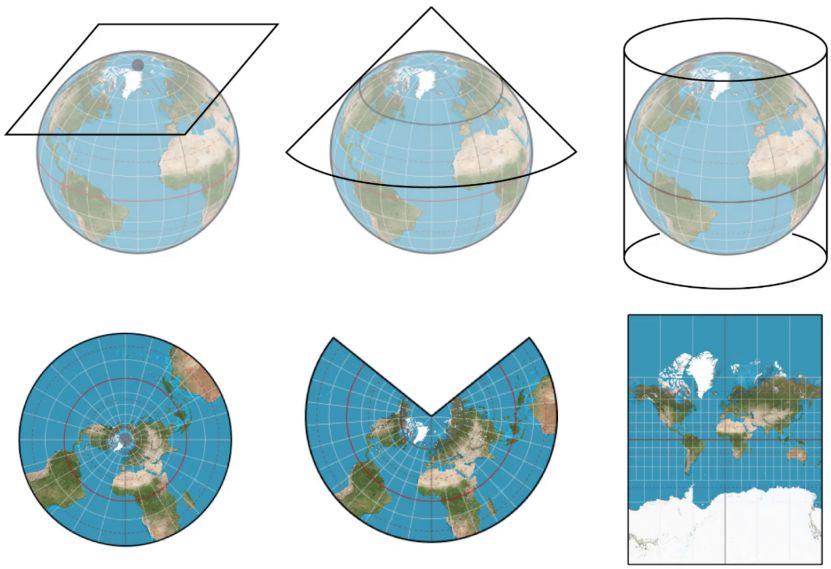
1. How should we generate background points if no true absences are confirmed? Which approach to generating a background area from which to draw points is most reliable? What if a species does indeed occur at a generated background point, but just hasn't been recorded there yet?
2. How do we verify that all presence points are real?
3. How do we determine which variables to include in the SDM? How might predictions change if only temperature, or temperature + rainfall + disturbance, or some other combination are used?
4. How do we assess the impact of adding more training points? What if there are very few presence points available?
5. How accurate are predictions based on broad variables like temperature and rainfall, which do not capture microhabitats?
6. Should we be considering the thermal physiology of the target species in the SDMs? How might the species' ability to adapt to new environments confound predictions?
7. How reliable are future climate models, considering the vast variety of potential scenarios?

Keep a list of other questions that pop up as you progress through the module.

5 Map projections

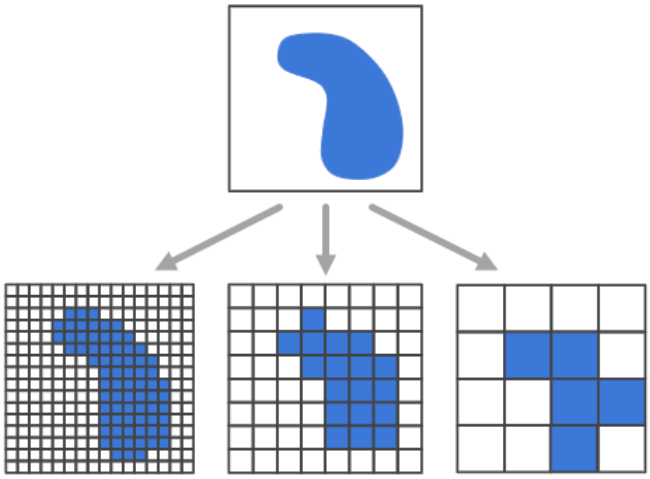
Because the Earth is a three-dimensional (approximately spherical) surface, representing it on a flat, two-dimensional raster is not straightforward. To deal with this, we use map projections, which transform the curved surface of the Earth into two dimensions while trying to minimise distortion. Spatial data are typically referenced using a global coordinate system such as WGS84 (World Geodetic System 1984), which defines locations in terms of latitude and longitude on an ellipsoidal model of the Earth. When we want to display these data on a flat map, a projection must be applied, and this process inevitably introduces some degree of distortion.

When running SDM pipelines, it is essential to ensure that all raster layers share the same extent, resolution, and coordinate reference system (CRS). Without this alignment, grid cells will not correspond spatially, leading to incorrect environmental values being associated with species occurrences.

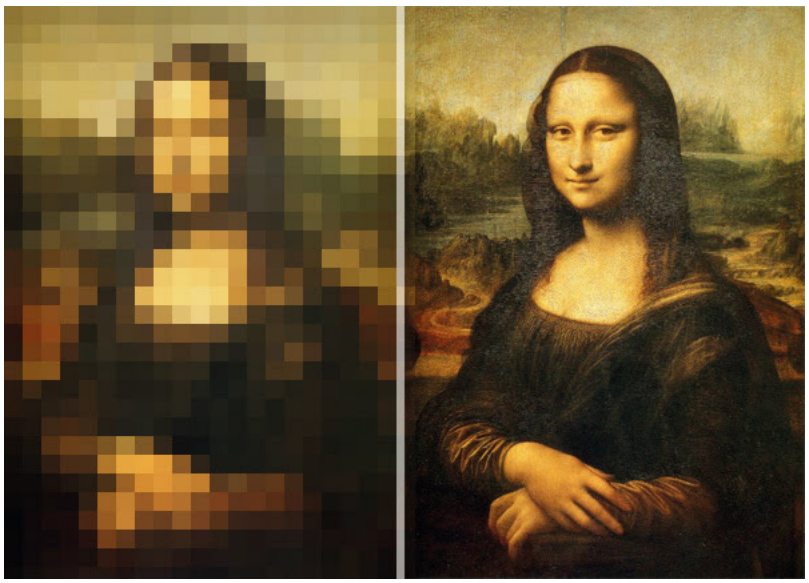


6 Raster resolution

The resolution of a raster refers to the size of each grid cell (i.e. the spatial grain). Higher resolution rasters have smaller cells and therefore capture finer spatial detail, while lower resolution rasters have larger cells and represent the landscape more coarsely.



The same concept applies to image pixels, which you will be more familiar with:



7 Readings

Below are readings assigned for each week to help build your understanding of the uses and limitations of SDMs. While going through them, keep the overall SDM workflow in mind and think about which part of the process the criticisms in the review articles are targeting:

data → input variables (=predictors) → model → evaluation → projection

By the Thursday of each week, you will be told which of the assigned readings will be used for a class discussion the following day. Each Friday, find a new partner to discuss the selected paper with. Chat about what you found interesting, what was confusing, and what score you would give the paper out of ten. We'll then discuss some key points as a class.

Week 1

1. Not as good as they seem: the importance of concepts in species distribution modelling (Jiménez-Valverde et al., 2008)
2. Species Distribution Models: Ecological Explanation and Prediction Across Space and Time (Elith and Leathwick, 2009)
3. Species Distribution Modeling (Jennifer Miller, 2010)
4. New trends in species distribution modelling (Zimmermann et al., 2010)
5. Top ten hazards to avoid when modeling species distributions: a didactic guide of assumptions, problems, and recommendations (Soley-Guardia et al., 2023)

Week 2

1. A practical guide to MaxEnt for modeling species' distributions: what it does, and why inputs and settings matter (Merow et al., 2013)
2. Oh the places they'll go: improving species distribution modelling for invasive forest pests in an uncertain world (Srivastava et al., 2020)
3. Species distribution models have limited spatial transferability for invasive species (Liu et al., 2020)
4. The dos and don'ts for predicting invasion dynamics with species distribution models (Hui, 2023)

Week 3

1. Improving species distribution models for climate change studies: variable selection and scale (Austin and van Niel, 2010)
2. Achieving higher standards in species distribution modeling by leveraging the diversity of available software (Kass et al., 2024)
3. Diversity Loss and Homogenization in Snake Communities in a Deforestation Hotspot (Cabral et al., 2025)
4. Spatial prediction of *Melanoplus differentialis* using an ensemble of multiple species distribution models (Yoon et al., 2025)

Week 4

1. Is my species distribution model fit for purpose? Matching data and models to applications (Guillera-Arroita et al., 2015)
2. The overlapping global distribution of dengue, chikungunya, Zika and yellow fever (Lim et al., 2025)
3. Flexible methods for species distribution modeling with small samples (Maitner et al,. 2026)

8 Interpretation exercise: hypothetical housing development

Scenario

A proposed housing development is planned in a region containing an endangered frog endemic to South Africa. An ensemble SDM has been developed (averaging the output from MaxEnt and Random Forest algorithms) to estimate the habitat suitability for the frog.

The development site falls within a region of moderate-to-high predicted suitability (approx. $P = 0.72$).

Only 8 confirmed presence records were available for model construction, and sampling is spatially biased toward accessible areas.

Model Performance Summary

Table 1: Summary of model performance metrics for MaxEnt, Random Forest, and the ensemble species distribution model, including discrimination ability (AUC), threshold-dependent accuracy (TSS and Kappa), and classification performance (sensitivity and specificity). Higher values indicate improved model performance for each metric.

Metric	MaxEnt	Random Forest	Ensemble
AUC	0.91	0.96	0.94
TSS	0.68	0.74	0.72
Kappa	0.55	0.63	0.60
Sensitivity	0.75	0.88	0.82
Specificity	0.81	0.85	0.83

Map Interpretation Task

You are provided with maps showing:

- Current climate suitability
- Future climate suitability
- Change in suitability (Future – Current)
- Species presence points
- The proposed housing development area

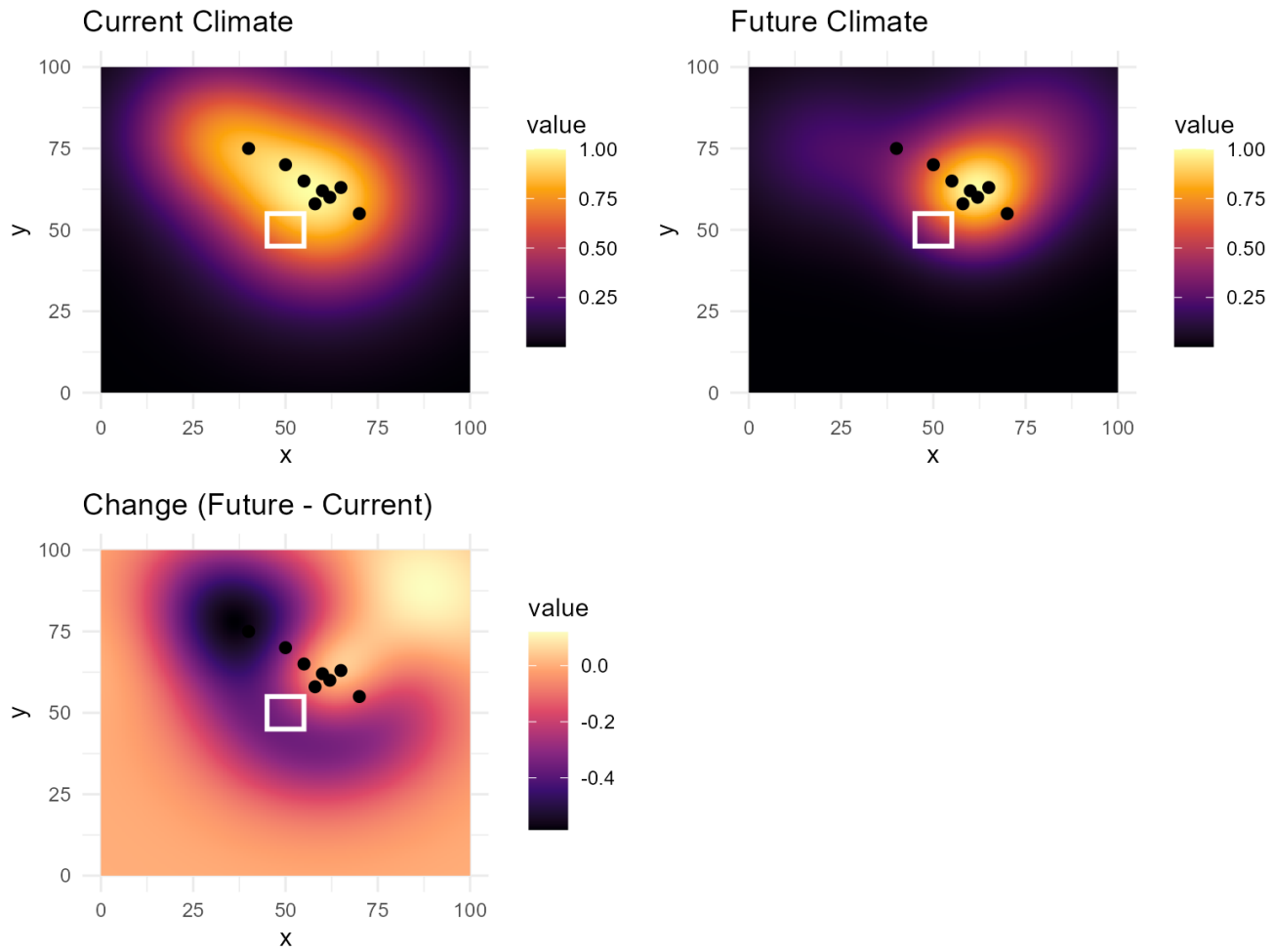


Figure 2: SDM outputs showing predictions for the current and future climate, and a raster showing the difference (i.e. estimated change over time). Black dots are the eight recorded presence points for the frog species, and the white rectangle is the location of the proposed housing development.

Consider:

- Where does the species currently occur relative to predicted suitability?
- How does suitability change under future climate conditions?
- Does the development site fall in stable, increasing, or decreasing suitability areas?

Role-Based Discussion

Divide into three groups, and run a mock debate. There should be developers who want to push for the housing construction to begin, conservation scientists who want to prevent the development, and regulators who make the final call. Although you are not experts yet in SDMs, use your logic and intuition, and play devil's advocate where necessary. Below are some prompts to get you thinking:

Developers

- What are the flaws in the evidence presented?
- Which metric or result supports your position most strongly?
- What level of uncertainty is acceptable for proceeding?

Conservation Scientists

- What possible ecological risks do the maps show?
- Which is more important: avoiding false positives or false negatives? (**False positive:** the model predicts suitable habitat where the species does not actually occur. **False negative:** the model predicts unsuitable habitat where the species actually occurs.)
- What arguments support blocking the development?

Regulators

- Is the evidence sufficient for a decision?
- Should precaution take preference over model performance metrics?
- What additional information would improve confidence in the predictions?

9 Raster exercise: Foundations of SDMs

This activity provides a hands-on introduction to the input data used in SDMs and how environmental information is extracted from raster layers.

You will be provided with four transparent plastic sheets. These represent a simplified geographic area divided into a 4 × 4 grid (columns A–D and rows 1–4). Each cell corresponds to a spatial location.

Three sheets contain environmental raster layers: mean temperature (Table 2), mean rainfall (Table 3), and elevation (Table 4) for the study area. The fourth sheet represents GPS presence and absence points for a taxon of interest (Fig. 3).

Here, each environmental sheet represents a **raster layer**, and together they form a **raster stack**.

The grids should show:

Table 2: Temperature raster (°C)

	A	B	C	D
1	18	19	22	25
2	17	18	23	26
3	16	17	24	27
4	15	16	25	28

Table 3: Rainfall raster (mm)

	A	B	C	D
1	900	850	500	300
2	950	800	450	250
3	1000	750	400	200
4	1050	700	350	150

Table 4: Elevation raster (m)

	A	B	C	D
1	200	300	800	1200
2	250	350	900	1300
3	300	400	1000	1400
4	350	450	1100	1500

And the GPS grid should resemble:

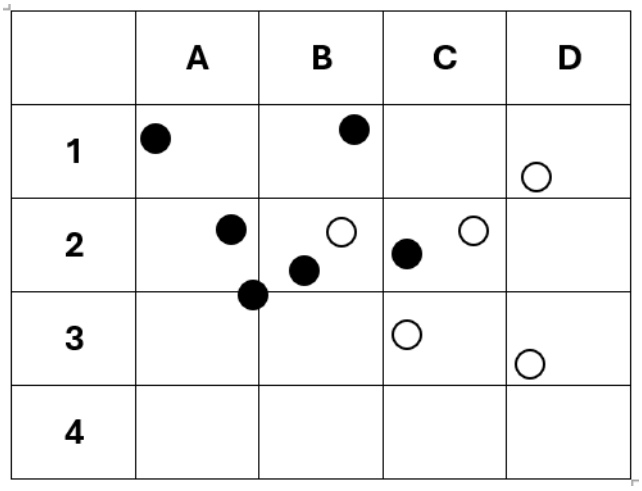


Figure 3: GPS grid, showing presences as black dots and absences as filled white dots.

Your tasks are to:

1. Create a table in Excel where you record each GPS point (presences and absences), and extract the environmental values (temperature, rainfall, and elevation) at each location. If a point falls within a single grid cell, record that cell value. If a point overlaps multiple grid cells, calculate the mean of those cell values.
   The table should look like this:

   Table 5: Extract environmental values at each point (fill in the blank cells).

Point ID	Presence	Temp (°C)	Rain (mm)	Elev (m)
A1	1	18	900	200
A2	1			
B1	1			
B2 (presence)	1			
B2 (background)	0			
Intersect	1			
C2 (presence)	1			
C2 (background)	0			
C3	0			
D1	0			
D3	0			

2. Read your Excel data table into R.

```r
# Reminder of reading files into R
# a standard CSV file
data = read.csv("path/to/your_file.csv")
# an xlsx file
library(readxl)
data = readxl::read_excel("path/to/your_file.xlsx", sheet = "
   sheet1")
# a txt file
library(readr)
readr::read_file("path/to/your_file.txt")
```

3. Fit a binomial GLM in R (logistic regression), where presence (0/1) is modelled as a function of temperature, rainfall, and elevation:

$$\text{presence} \sim \text{temp} + \text{rain} + \text{elev} \tag{1}$$

Are any of the predictor variables statistically significant? If not, provide a brief explanation based on sample size, spatial resolution, or data structure.

```r
# Reminder of the GLM format in R
model = glm(presence ~ temp + rain + elev,
            data = df,
            family = binomial)
```

The model you have just created can be written like this:

$$\text{presence} \sim \text{Binomial}(p) \tag{2}$$

where p is the probability of species presence.
For those interested in the maths behind it, the final predictive equation is derived from these steps:

The GLM assumes a linear relationship between predictors and the log-odds of presence (η is the Greek letter "eta", often used in GLM notation):

$$\eta = \beta_0 + \beta_1 \cdot \text{temp} + \beta_2 \cdot \text{rain} + \beta_3 \cdot \text{elev} \tag{3}$$

In this model, η can be interpreted as the log-odds of species presence, before it is transformed into a probability between 0 and 1.
If we substitute the estimated model coefficients:

$$\eta = -52.71 + 1.627 \cdot \text{temp} + 0.03026 \cdot \text{rain} + 0.0003567 \cdot \text{elev} \tag{4}$$

The relationship between probability and the linear predictor is given by the logit link:

$$\log\left(\frac{p}{1-p}\right) = \eta \tag{5}$$

Substituting the linear predictor:

$$\log\left(\frac{p}{1-p}\right) = -52.71 + 1.627 \cdot \text{temp} + 0.03026 \cdot \text{rain} + 0.0003567 \cdot \text{elev} \tag{6}$$

To obtain the probability of presence, we apply the inverse-logit function:

$$p = \frac{1}{1+e^{-\eta}} \tag{7}$$

Substituting η, the final equation is:

$$p = \frac{1}{1+\exp\left[-\left(-52.71 + 1.627 \cdot \text{temp} + 0.03026 \cdot \text{rain} + 0.0003567 \cdot \text{elev}\right)\right]} \tag{8}$$

4. Use the model to predict the probability of presence for each GPS point using the `predict()` function. Add these values to your original data frame as a column called `pred`.

```
df$pred = predict(model, newdata = df, type = "response")
```

5. Create a confusion matrix comparing observed and predicted outcomes. A threshold of 0.5 is used here for simplicity to convert probabilities into binary classes (presence/absence). In practice, threshold selection usually depends on various model evaluation metrics.

```
df$pred_class = ifelse(df$pred > 0.5, 1, 0)

conf_matrix = table(
  Predicted = df$pred_class,
  Observed  = df$presence
)

conf_matrix
```

6. Using the confusion matrix, calculate the model's accuracy, sensitivity, and precision using the formulae below. The confusion matrix is defined with rows as Predicted and

columns as Observed values.

- True positive (TP): Predicted = 1, Observed = 1
- True negative (TN): Predicted = 0, Observed = 0
- False positive (FP): Predicted = 1, Observed = 0
- False negative (FN): Predicted = 0, Observed = 1

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad \text{(overall correct classification rate)}$$

$$\text{Sensitivity} = \frac{TP}{TP + FN} \quad \text{(ability to detect presences)}$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad \text{(reliability of predicted presences)}$$

Accuracy measures the proportion of **all predictions** that are correct. I.e. Out of everything, how many predictions did the model get right?

Sensitivity (also called "recall" or "true positive rate") focuses only on **actual presences**. I.e. of all the real presences, how many did the model detect?

Precision focuses only on **predicted presences**. I.e. of all predicted presences, how many are actually correct?

7. How well do you think the model performed? Note that relying on a single accuracy metric can be misleading, and sensitivity is often more useful for assessing a model's performance.
8. Create a raster layer for each environmental variable (temperature, rainfall, and elevation). Combine these layers into a raster stack. Using the binomial GLM, predict the probability of presence for each cell in the raster stack, and plot the resulting surface.
9. Create a binary map by applying a threshold of 0.5 to the predicted probabilities. Cells with values greater than 0.5 are classified as suitable, and those below as unsuitable. Which parts of the geographic area are most suitable for this taxon? Describe the spatial pattern in relation to the environmental gradients (you can plot each individual raster to look at the patterns).

```r
# Create the temperature df
temp_r = matrix(c(
  18, 19, 22, 25,
  17, 18, 23, 26,
  16, 17, 24, 27,
  15, 16, 25, 28
), nrow = 4, byrow = TRUE)

# Create the rainfall df
rain_r = matrix(c(
  900, 850, 500, 300,
  950, 800, 450, 250,
  1000, 750, 400, 200,
  1050, 700, 350, 150
), nrow = 4, byrow = TRUE)

# create the elevation df
elev_r = matrix(c(
  200, 300, 800, 1200,
  250, 350, 900, 1300,
  300, 400, 1000, 1400,
  350, 450, 1100, 1500
), nrow = 4, byrow = TRUE)

# convert each to raster layers here

# assign an appropriate name to each layer using names(layer) = "
   name"

# combine the layers into a single raster stack called env
env = c(layer1, layer2, layer3)

# predict suitability across the raster stack, using the binary
   GLM created earlier
suitability = predict(env, model, type = "response")
```

```
    # plot the predictions
terra::plot(suitability, main = "Habitat Suitability (GLM SDM)")

# threshold the predictions
habitat_binary = suitability > 0.5

terra::plot(habitat_binary,
            main = "Binary Habitat Suitability (Threshold = 0.5)"
               ,
            col = c("#d9d9d9", "#238b45"))
```

You should see the following two prediction grids:

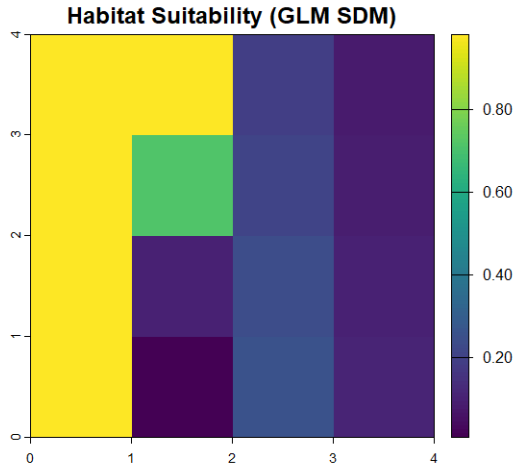


Figure 4: Continuous prediction grid

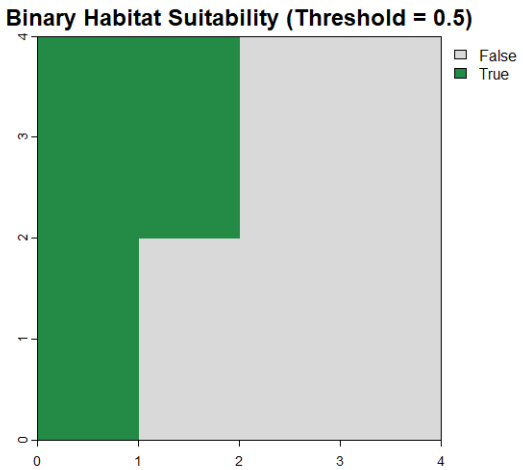


Figure 5: Binary prediction grid, thresholded to show suitability scores > 0.5.

TO HAND IN:

Submit the following documents:

1. The Excel file with the full data you compiled from the physical plastic sheets that represented a raster stack.
2. A single R script with all your code for this exercise. Include comments explaining what each section does.
3. A Word document (1 page max) where you record the binomial GLM's accuracy, sensitivity, and precision metrics, and the final two prediction grids (continuous and thresholded binary). Write a short paragraph explaining what you think would happen in the binary prediction grid if (1) the original raster layers (temp, rain, elev) had a lower resolution, and (2) the applied suitability threshold was lowered to < 0.5.

In this exercise, we have used presence and absence points, where we assume that the absence points are true, recorded areas where the taxon does not occur. In reality, we usually do not have true absence data. How do we know that the species was not actually present but simply not detected at the time of observation?
This is why tools like MaxEnt are very popular, because they are **presence-only** models. Instead of true absences, points are sampled from a background area. We call these "**background points**". Rather than representing real absences, they represent the environmental conditions available across the landscape, including locations where the species may or may not occur, but where it was not recorded. Presence-only models code background points as zeroes, but do not treat them as true absences.

10 Module Assignment [100 marks]

Assignment: Using all available GPS data on GBIF for the Argentine ant (*Linepithema humile*), create an ensemble species distribution model (SDM) and project it onto southern Africa.

Tasks

1. **Data acquisition and cleaning:**
   - Download all available occurrence records from GBIF.
   - Check for duplicates, coordinate errors, and other anomalies.
2. **Environmental variables:**
   - Start with all 19 WorldClim variables.
   - Assess collinearity and select a suitable subset.
3. **Background points:**
   - Create background points using the Köppen-Geiger approach from the ACP tutorial or another method of your choice.
   - Ensure reproducibility by setting a random seed.
4. **Ensemble modeling:**
   - Use the following algorithms: MAXNET, Random Forest (RF), and Generalised Linear Model (GLM). If desired, include a fourth method and document it.
   - For MAXNET, use default settings: feature class (FC) = LQH, regularisation multiplier (RM) = 1.
   - Record performance metrics for each model (ROC/AUC, TSS, Kappa, sensitivity, specificity).
5. **Projection and mapping:**
   - Project the ensemble model onto southern Africa (include South Africa, Namibia, Botswana, Zimbabwe, eSwatini, Lesotho, Mozambique, Zambia, and Malawi).
   - Produce a map of predicted suitability and a second version overlaying observed occurrences.
   - Include a scale bar, legend, and north arrow.
6. **Data availability and reproducibility:**
   - Create a GitHub repository (public) and store your R scripts there. Include an informative README file. In your write-up, provide the GitHub repository link.

Submission requirements

- A **single zipped folder** containing:

- Clean **R code** (`.R` files only). Use comments (`#`) to clearly separate sections. Example structure:

```
#################
# GET DATA
#################
your.code.here

#================
# CLEAN DATA
#================
more.code.here

#----------------
# COLLINEARITY
#----------------
more.code.here
```

- A **Word document** summarising your analysis, following the structure of a scientific article:
  * Introduction (keep short)
  * Materials and Methods, with headings that match Tasks 1 - 6
  * Results
  * Discussion
  * References
- Include a table with model performance metrics and your final prediction map.
- Discuss which parts of South Africa are most climatically suitable and whether there are potentially suitable areas without records.

Additional guidance

- Clearly explain any assumptions or decisions made during modelling.
- Briefly discuss model limitations and uncertainty.
- Think about possible invasion pathways and suitable areas that could serve as corridors.
- Ensure reproducibility by setting random seeds where applicable.
- Name your files and folders clearly to avoid confusion.

Mark breakdown

Component	Weight	Description
Analysis accuracy	60%	Correctness of data cleaning, variable selection, ensemble modeling, and interpretation of results.
R code and GitHub curation	30%	Clear, well-commented code; organised GitHub repository; reproducible workflow.
Report and figure presentation	10%	Neat, well-structured report with clear figures and tables.

11 Terminology

Term	Definition
Core Concepts	
Ecological Niche	The set of environmental conditions under which a species can persist.
Fundamental Niche	The full range of environmental conditions a species can tolerate in the absence of biotic constraints.
Habitat Suitability	A continuous index (often 0–1) representing how suitable a location is for a species.
Realised Niche	The subset of the fundamental niche constrained by biotic interactions and dispersal limitations.
Species Distribution Model (SDM)	A model that relates species occurrence or abundance data to environmental predictors to estimate habitat suitability or geographic distributions.
Data Types	
Absence Data	Locations where a species is confirmed not to occur.
Background Data	Environmental samples used in presence-only models to contrast with occurrence data.
Presence Data	Locations where a species has been observed.
Pseudo-absence Data	Artificially generated absence points used when true absences are unavailable.
Sampling Bias	Non-random spatial bias in occurrence data due to uneven survey effort.
Environmental Data & Predictors	
Collinearity	High correlation among predictor variables, which can affect model performance.
Extent	The geographic area covered by the dataset.
Predictor Variables	Environmental variables used to explain species distributions (e.g., climate, soil).
Raster Data	Gridded spatial data where each cell contains a value.
Resolution	The spatial grain size of raster data.
Variance Inflation Factor (VIF)	A metric used to quantify multicollinearity among predictors.
Model Types	

Term	Definition
Artificial Neural Network (ANN)	A machine learning model inspired by biological neural systems, consisting of interconnected nodes (neurons) organized in layers.
Boosted Regression Trees (BRT)	A tree-based method that builds models sequentially to improve performance.
Ensemble Modelling	Combining predictions from multiple models to improve robustness.
Generalized Additive Model (GAM)	A flexible extension of GLMs that allows non-linear relationships via smooth functions.
Generalized Linear Model (GLM)	A regression-based model that can handle different response distributions via link functions.
MaxEnt	A presence-background modelling approach based on maximum entropy principles.
Random Forest	An ensemble machine learning method using multiple decision trees.
Model Calibration & Tuning	
Cross-validation	A resampling method used to assess model generality.
Feature Class	Define the types of transformations applied to predictor variables (e.g., linear, quadratic, hinge, product). They control model flexibility and complexity by determining how environmental relationships are represented.
Hyperparameters	Model settings that are specified prior to training and not learned from the data.
Regularisation	A technique to penalize model complexity and reduce overfitting.
Training Data	Data used to fit the model.
Validation Data	Independent data used to evaluate model performance.
Model Evaluation	
AUC	Area Under the Receiver Operating Characteristic Curve; measures model discrimination ability.
Kappa	A statistic measuring agreement between observed and predicted values.
Sensitivity	Proportion of presences correctly predicted.
Specificity	Proportion of absences correctly predicted.
Threshold	A cutoff used to convert continuous predictions into binary outcomes.
TSS	True Skill Statistic; balances sensitivity and specificity.

Term	Definition
Predictions & Transferability	
Clamping	Restricting predictions to the range of training data values.
Extrapolation	Predicting outside the range of training data.
Projection	Applying a model to new environmental data (e.g., future climate scenarios).
Transferability	The ability of a model to perform well in new contexts.
Spatial & Ecological Considerations	
Biotic Interactions	Interactions among species (e.g., competition, predation).
Dispersal Limitation	Constraints on species movement affecting distributions.
Niche Conservatism	The tendency of species to retain ancestral ecological traits.
Spatial Autocorrelation	The non-independence of observations due to spatial proximity.
Spatial Cross-validation	Cross-validation that accounts for spatial autocorrelation.
Workflow & Best Practices	
Data Wrangling	The process of cleaning and preparing data for analysis.
GIGO	"Garbage In, Garbage Out"; poor input data leads to poor model outputs.
Overfitting	When a model captures noise rather than true signal.
Pipeline	A structured sequence of analytical steps.
Reproducible Workflow	An analysis that can be rerun with identical results.
Version Control	Tracking changes in code and data over time.